ITAI 1371 ? Module 10

INDIVIDUAL CONTRIBUTION JOURNAL

Alexander DeBusk

Unsupervised Learning

L10 – Alexander DeBusk – Contributions

- Completed the coding exercises for K-Means clustering, the Elbow Method, and Principal Component Analysis (PCA).
- Ran and reviewed the code, corrected an issue with the Seaborn import, and checked the resulting visualizations.
- Completed the Knowledge Check questions covering supervised and unsupervised learning, selecting the number of clusters, and PCA.
- Saved all deliverables using the appropriate file-naming conventions and submitted them on time.

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INDIVIDUAL CONTRIBUTION JOURNAL

Shareefah Adeola

Unsupervised Learning

CONTRIBUTION JOURNAL: **Module 10: Unsupervised Learning**

ITAI 1371

Student: Shareefah Adeola

Date: November 4, 2026

Summary of My Contribution

I completed the Module 10 Unsupervised Learning lab by working through the Python coding activities, reviewing the visualizations, correcting errors, and answering the Knowledge Check questions.

Work Completed

- Generated and visualized 300 synthetic data points, using different colors to show the four groups.
- Applied the Elbow Method by calculating and plotting inertia for k values from 1 to 10; selected k = 4.
- Ran the K-Means clustering with four clusters and displayed the cluster centers.
- Applied PCA to the Iris dataset, reducing four features to two principal components.
- Confirmed that the two PCA components explained approximately 97.77% of the total variance.
- Completed the three Knowledge Check responses and reviewed the notebook outputs.

Result

Deliverable	Outcome
Synthetic dataset	300 observations generated and displayed in four colored groups.

Elbow Method	Inertia calculated for k = 1 to 10; k = 4 selected for the next task.
K-Means model	Four clusters visualized with their centroids.
PCA	Four Iris features reduced to two principal components.
Explained variance	Approximately 97.77% of variance is retained by the two components.
Written work	Three Knowledge Check responses prepared, and notebook review document created.

Challenges and Learning

I corrected code that had been placed in the wrong notebook section and fixed an indentation error in the PCA code. I also learned how to use cluster labels to display data points in different colors. This lab improved my understanding of K-Means clustering, the Elbow Method, PCA, and interpreting machine-learning visualizations.

Deliverable

The completed Jupyter Notebook contains the code, visualizations, results, and Knowledge Check answers. I also prepared supporting review and reflection documents.

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INDIVIDUAL CONTRIBUTION JOURNAL

Yilin Leng

Unsupervised Learning

Contribution

In this Unsupervised Learning experiment, I independently completed the coding, testing, and visualization of the K Means clustering algorithm and Principal Component Analysis (PCA). I used sklearn.cluster.KMeans to test different cluster numbers from 1 to 10 and calculate the inertia for each model. By using the Elbow Method, I identified K = 4 as the best number of clusters. I also created visualizations of the data points, cluster labels, and four centroids using red X markers. For PCA, I used sklearn.decomposition.PCA to reduce the original four-dimensional Iris dataset to two principal components. The results showed that the first two components retained 97.77% of the total variance. I also used Pandas and Seaborn to create a scatter plot showing the distribution of the three Iris species. In addition, I completed the theoretical questions about supervised learning, unsupervised learning, the Elbow Method, and PCA. I explained why using too many clusters is not a good choice and how PCA can reduce the number of features while keeping most of the important information. Overall, I completed the main coding tasks, checked the results, created the visualizations, and finished the related analysis independently.